

Addendum: A neural circuit for wind-guided olfactory navigation

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Andrew M. M. Matheson , Aaron J. Lanz , Ashley M. Medina, Al M. Licata,
 Timothy A. Currier , Mubarak H. Syed & Katherine I. Nagel 

In our published Article, we identified a Gal4 line (VT062617) that we believed labeled hΔC interneurons of the fan-shaped body. We showed that fan-shaped body interneurons labeled by this line responded to odor in a wind-direction-dependent manner and that both activation and silencing of neurons labeled by this line had effects on fly navigation behavior. Following the publication of the Article, we performed additional experiments to label single neurons within this line, now published in Hamid et al. 2024¹. These experiments suggest that this line additionally or perhaps predominantly labels a different fan-shaped body interneuron type known as hΔK, which is characterized by a projection to the ellipsoid body. Thus, the sensory responses (Figure 5E–I) and behavioral effects (Figure 6) that we described in the original Article may arise from hΔK rather than hΔC interneurons. In contrast to hΔC, which receives direct projections from the olfactory neuron FB5AB and from wind-sensitive PFNa neurons, as shown in Figure 5B, C of the original Article, hΔK interneurons do not receive direct projections from these two neuron types, although they do receive indirect inputs from them. Thus, the sensory responses shown in Figure 5E–I in line VT062617 may arise through indirect pathways.

Reference

1. Hamid et al. The conserved RNA-binding protein Imp is required for the specification and function of olfactory navigation circuitry in *Drosophila*. *Curr. Biol.* **34**, 473–488 (2024).

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